

CellRank: Directed Single-cell Fate Mapping

01. Paper Information

Paper	CellRank for directed single-cell fate mapping
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02. At a Glance

No.	Aspect	Reading note
01	Question	How can we infer directed and probabilistic cell-fate trajectories from snapshot scRNA-seq data when the biological direction is not known in advance?
02	Input	A single-cell count matrix together with RNA velocity vectors. In principle, the framework can also use other directional vector fields such as metabolic labeling or experimental time.
03	Model	CellRank builds a directed Markov chain on a cell-cell similarity graph, biases transitions by RNA velocity, coarse-grains the chain into macrostates with GPCCA, and computes absorption probabilities toward terminal states.
04	Output	Initial, intermediate, and terminal macrostates; fate probabilities for each cell; lineage-specific gene-expression trends; and candidate lineage drivers.
05	Main claim	By combining similarity-based trajectory inference with RNA velocity, CellRank yields robust, global, and probabilistic fate maps in development, reprogramming, and regeneration.
06	Why it matters	The key conceptual move is to go beyond local arrows: CellRank translates noisy local directional evidence into a global and interpretable map of likely terminal fates.

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1 One-sentence summary

Core summary

CellRank converts local RNA velocity and cell-cell similarity into a directed Markov process, then uses coarse-grained state dynamics to identify initial and terminal populations and compute each cell's probability of reaching every terminal fate.

2 Why this paper matters

Single-cell RNA sequencing captures many cells at different states, but each cell is usually measured only once. This means that we observe a static snapshot of a dynamic process. Classical pseudotime methods can order cells along a trajectory, but they usually require prior knowledge about the starting point and direction of the process. RNA velocity provides a local direction by estimating whether genes are being up- or down-regulated from spliced and unspliced mRNA, but raw velocity vectors are noisy and are often interpreted only on two-dimensional embeddings.

CellRank is important because it bridges these two ideas. It keeps the robustness of neighborhood-based trajectory inference while injecting directionality from RNA velocity. More importantly, it does not stop at local arrows. It aggregates local transition signals into a global probabilistic fate map.

Key takeaway

The main conceptual shift is from **local velocity arrows** to **global fate probabilities**. Instead of asking only where a cell may move next, CellRank asks which terminal population a cell is likely to reach after many small transitions on the phenotypic manifold.

3 Biological and computational background

3.1 The biological problem

Many biological systems involve continuous state transitions: development, reprogramming, regeneration, and disease progression. These processes are often asynchronous, so a single scRNA-seq experiment may contain cells at many different transition stages. The challenge is to reconstruct a dynamic process from unordered snapshots.

There are three common sources of dynamic information:

- **Pseudotime:** orders cells by similarity, but usually needs an initial state or direction.
- **Lineage tracing:** provides clonal ancestry, but is mostly restricted to experimental systems where barcoding is possible.
- **RNA velocity:** estimates local transcriptional direction from spliced/unspliced ratios, but can be noisy and difficult to interpret globally.

Why CellRank is needed

Problem 3.1

A velocity vector tells us a local tendency, not a complete fate decision. A cell near a branching region may have multiple possible futures. Therefore, fate mapping should be probabilistic rather than deterministic.

3.2 The computational framing

CellRank formulates cell-state dynamics as a **Markov chain**. Each observed cell is a state. Edges connect transcriptionally similar cells, and edge weights represent transition probabilities. Directionality is introduced by comparing the RNA velocity vector of a cell with the expression difference vector toward each neighbor.

If cell i has velocity vector v_i and neighbor j differs by δ_{ij} in gene-expression space, a simplified transition score can be written as:

$$p_{ij} \propto \exp(\text{corr}(v_i, \delta_{ij})). \quad (1)$$

The actual transition matrix combines velocity-based directionality with expression similarity. The resulting transition matrix is high-dimensional, directed, and nonsymmetric.

Method note

The Markov formulation is the methodological core. It converts noisy local estimates into a global process. Once the process is built, terminal states and fate probabilities become questions about long-term transition behavior.

4 Method breakdown

4.1 Step 1: build a phenotypic neighborhood graph

CellRank first constructs a K-nearest-neighbor graph in the high-dimensional expression manifold. Each node is a cell, and edges connect cells with similar transcriptomic profiles. This step enforces the assumption that real cell-state transitions occur through small changes and transitional populations, rather than arbitrary jumps across expression space.

4.2 Step 2: direct the graph with RNA velocity

For each cell, CellRank uses its RNA velocity vector to bias transitions toward neighbors that lie in the predicted future direction. A neighbor aligned with the velocity vector receives higher transition probability. A neighbor opposite to the velocity vector receives lower probability.

Why not just use the UMAP arrows?

Important detail 4.1

UMAP or t-SNE arrows can be visually useful, but they are low-dimensional projections. They may distort global distances and over-smooth the true high-dimensional velocity field. CellRank computes transitions in gene-expression space rather than relying on the visual projection.

4.3 Step 3: coarse-grain the Markov chain into macrostates

The full cell-cell transition matrix is large and noisy. CellRank uses Generalized Perron Cluster Cluster Analysis (GPCCA) to summarize cells into **macrostates**: regions of the phenotypic manifold that cells are unlikely to leave quickly. This yields a smaller coarse-grained transition matrix among macrostates.

Macrostates are then interpreted as:

- **Terminal states:** high self-transition probability, measured by the stability index.
- **Initial states:** low probability of being revisited after leaving, measured through the coarse-grained stationary distribution.
- **Intermediate states:** remaining macrostates that lie along transition paths.

4.4 Step 4: compute fate probabilities

After terminal macrostates are identified, CellRank computes for each cell the probability of eventually reaching each terminal state. Mathematically, this is an absorption-probability problem on a Markov chain.

A useful block representation is:

$$P = \begin{pmatrix} Q & R \\ 0 & I \end{pmatrix}, \quad (2)$$

where Q describes transitions among transient states and R describes transitions from transient states to terminal states. The absorption probabilities can be obtained from:

$$B = (I - Q)^{-1}R. \quad (3)$$

Here, B_{ik} is the probability that cell i eventually reaches terminal fate k .

Key takeaway

Fate probability is the most reusable output of CellRank. It can be used to color cells, quantify lineage bias, define progenitor populations, and weight gene-expression trends along each lineage.

4.5 Step 5: infer lineage-specific gene programs

CellRank combines fate probability with pseudotime to visualize gene-expression dynamics along each lineage. Cells are not assigned hard labels to one branch. Instead, they contribute to multiple lineages according to their fate probabilities. This soft assignment is biologically meaningful because lineage commitment is gradual.

Genes whose expression correlates with fate probabilities are treated as candidate lineage drivers. Their expression trends can be ordered by pseudotime to reveal cascades of gene activation.

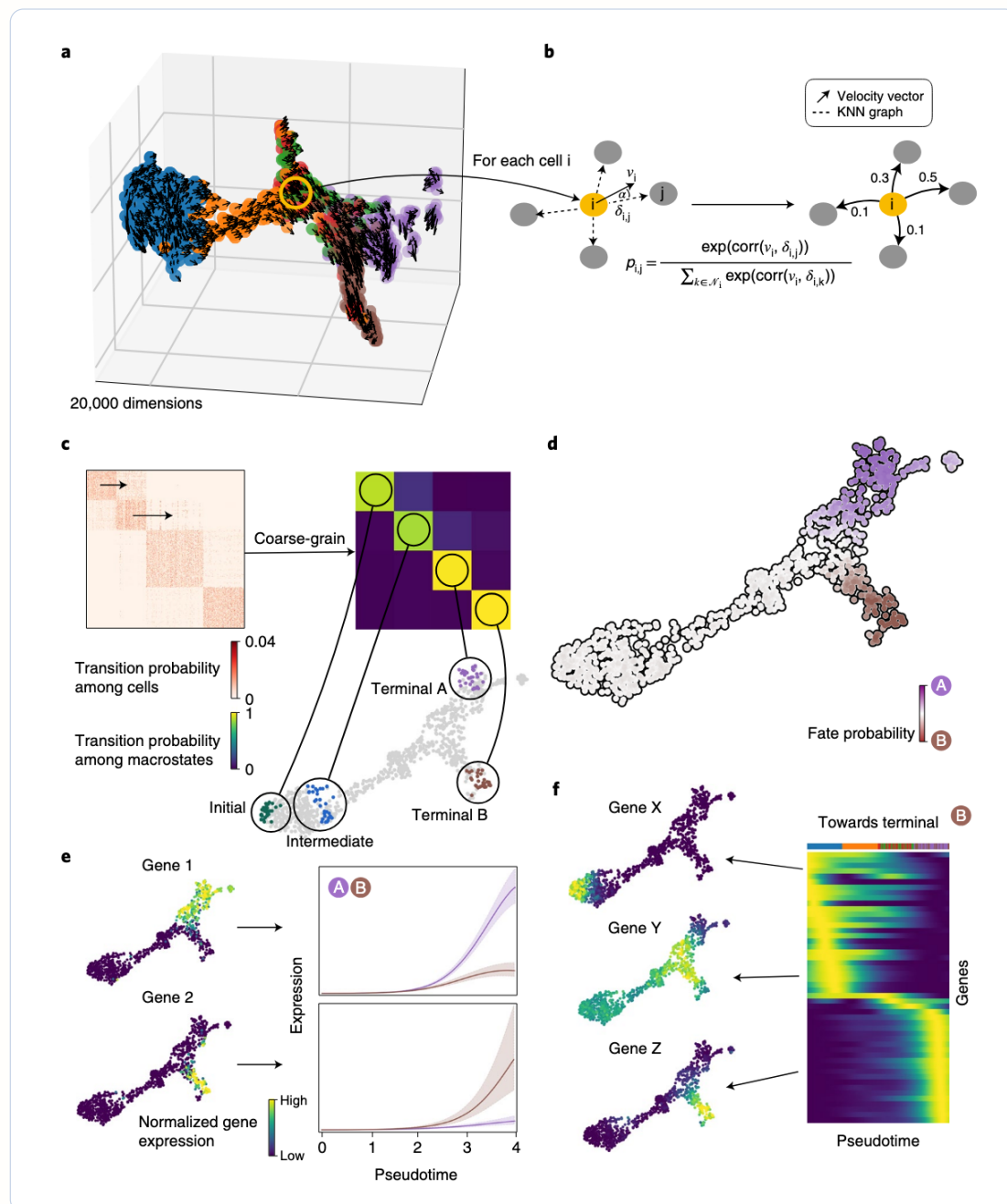


Figure 1: Main text Fig. 1a–f. Method overview of CellRank. Use this figure to show how RNA velocity and cell-cell similarity are converted into a directed transition matrix, macrostates, fate probabilities, and lineage-specific gene-expression trends.

How to read Fig. 1

Figure note 4.2

This figure is the best single summary of the method. Panel b explains the local transition rule, panel c explains coarse-graining, panel d explains fate probability, and panels e–f explain how fate probabilities support lineage-specific gene interpretation.

5 Main experiments and evidence

5.1 Pancreatic endocrine development

The first major experiment applies CellRank to murine pancreatic endocrine development at embryonic day E15.5. The biological trajectory starts from endocrine progenitors and branches toward alpha, beta, epsilon, and delta endocrine cell fates.

CellRank identifies macrostates corresponding to developmental stages, including Ngn3-low endocrine progenitors, Ngn3-high endocrine progenitors, Fev-positive intermediate states, and hormone-producing terminal states. The automatically identified terminal states include alpha, beta, and epsilon cells. Delta cells are detected as a relatively stable macrostate, but because they are rare, they are not automatically classified as terminal in the default setting.

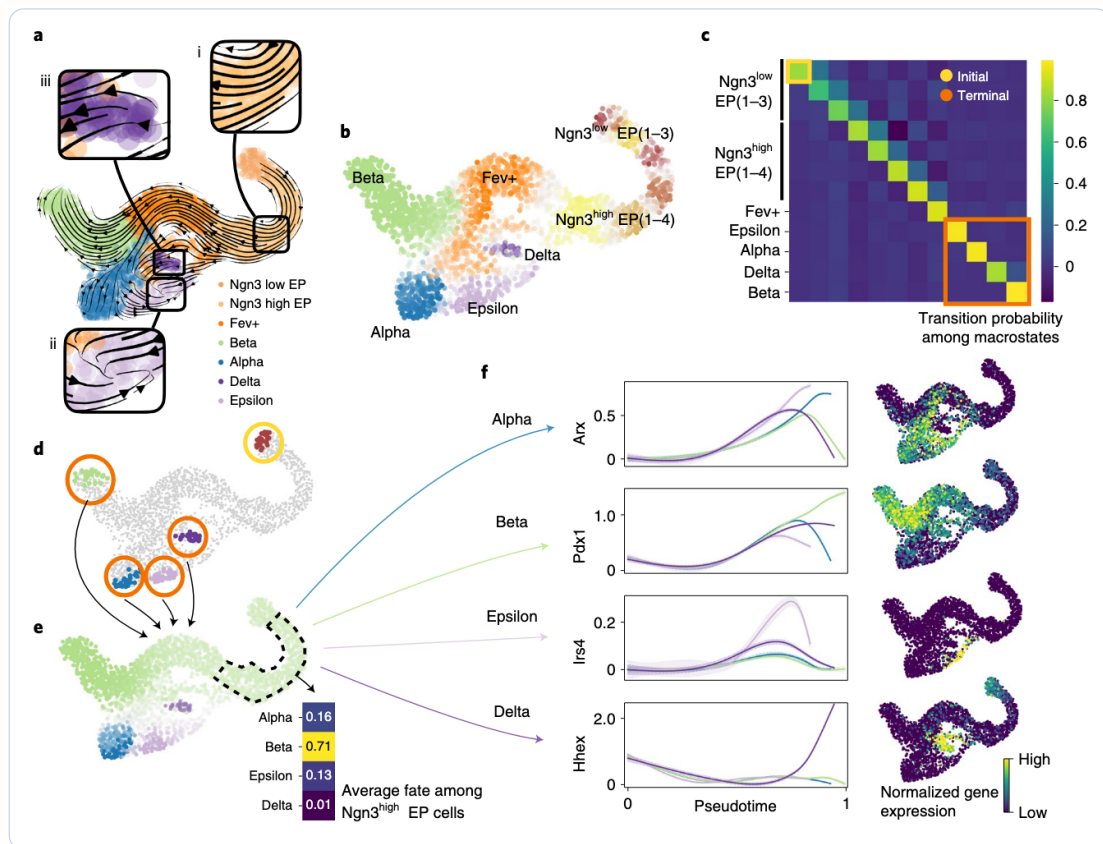


Figure 2: Main text Fig. 2a–f. Pancreatic endocrine development. CellRank recovers initial Ngn3-low endocrine progenitors, terminal endocrine fates, fate probabilities, and lineage-specific marker trends such as Arx, Pdx1, Irs4, and Hhex.

Interpretation

Figure note 5.1

This experiment demonstrates the complete CellRank workflow in a well-understood developmental system. The strongest evidence is not only that terminal states are recovered, but also that fate probabilities identify the beta lineage as dominant among Ngn3-high endocrine progenitors and recover expected lineage marker trends.

5.2 Delta-cell differentiation and candidate driver genes

The delta-cell analysis is particularly interesting because delta cells are rare and their developmental regulation is less well understood. The authors focus on fate probabilities toward the delta macrostate. Although projected

RNA velocity alone does not clearly reveal delta precursors, CellRank identifies a likely path through Fev-positive delta-precursor cells.

By correlating gene expression with delta fate probabilities, the paper recovers known delta-associated genes such as *Hhex*, *Cd24a*, and *Sst*. It also proposes candidate regulators including *Hadh*, *Isl1*, *Pkhd1*, *Map2k4*, *Msi1*, and *Nefl*.

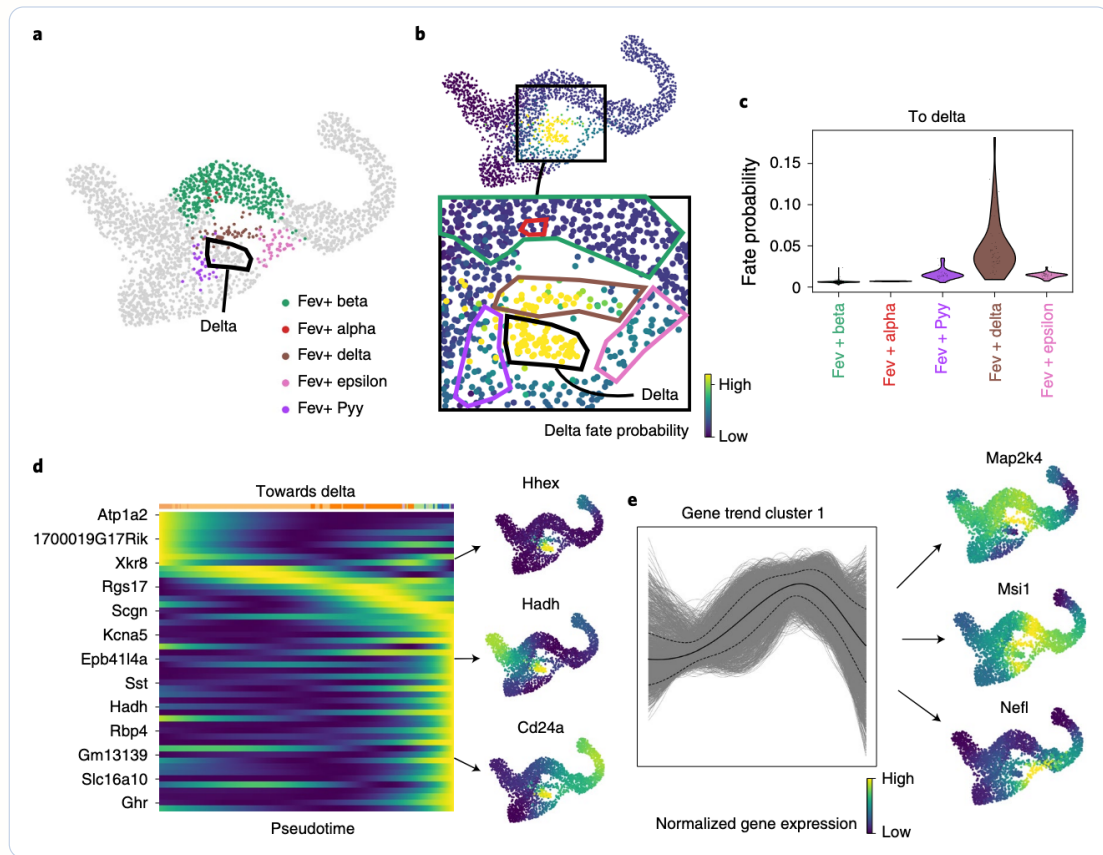


Figure 3: Main text Fig. 3a–e. Delta-cell differentiation analysis. CellRank uses delta fate probabilities to identify likely precursor cells and to prioritize candidate genes associated with delta-cell fate acquisition.

Why this result matters

Figure note 5.2

This result shows the interpretability value of fate probabilities. CellRank is not only a trajectory visualization method; it can produce a ranked list of lineage-associated genes and suggest new biological hypotheses.

5.3 Fibroblast-to-endoderm reprogramming

The second major setting is cellular reprogramming. Mouse embryonic fibroblasts are reprogrammed toward induced endoderm progenitors, but only a small fraction of cells succeed. The dataset includes CellTagging lineage-tracing information, which provides ground-truth labels for successful versus dead-end outcomes.

Projected RNA velocity does not clearly show a route toward successful reprogramming and can even suggest misleading transitions. In contrast, CellRank identifies both successful and dead-end macrostates. Fate probabilities toward the successful macrostate predict lineage-tracing-derived outcomes, with higher AUC at later time points.

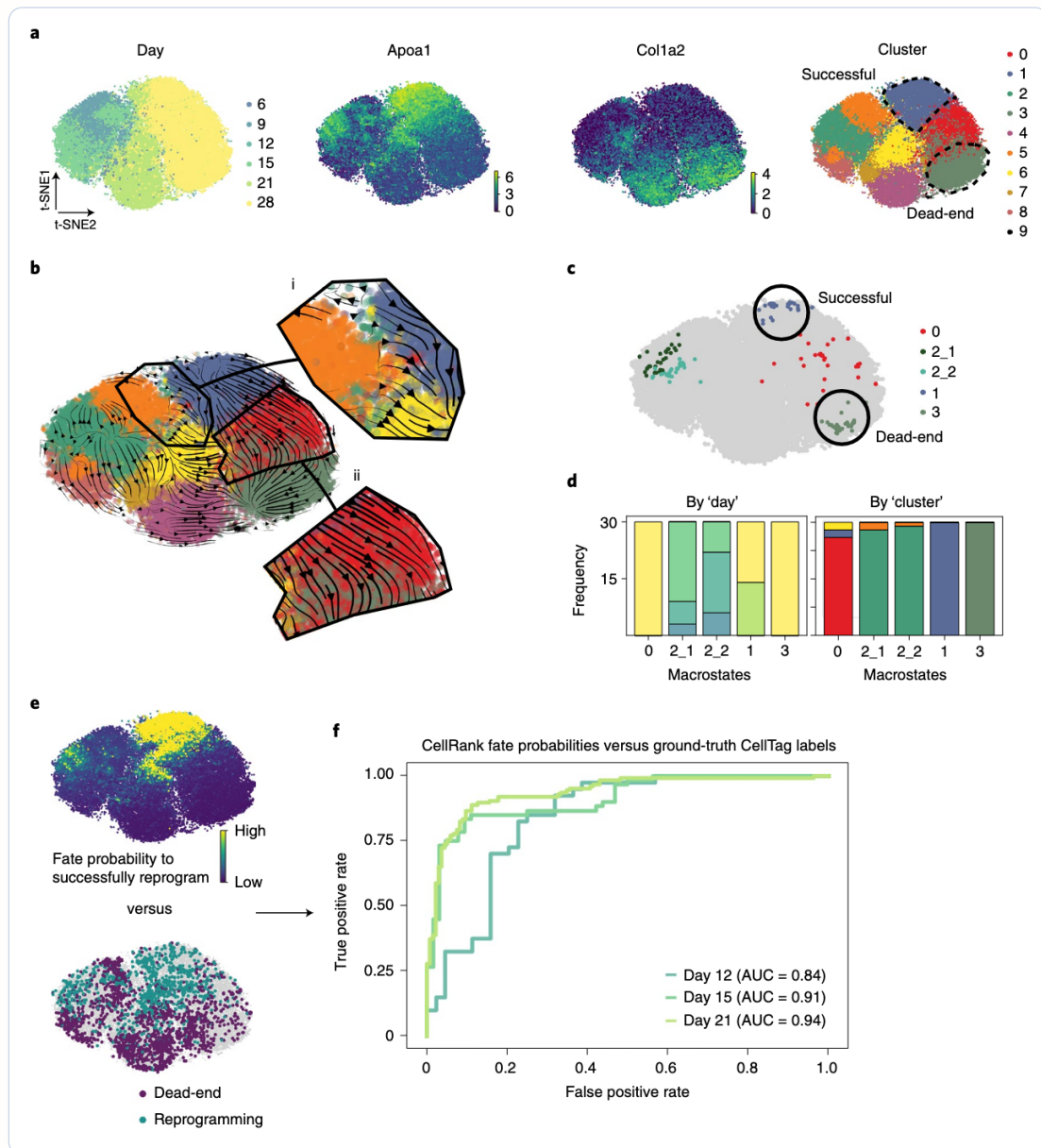


Figure 4: Main text Fig. 4a–f. Reprogramming outcome prediction. CellRank fate probabilities toward the successful macrostate agree with CellTagging lineage-tracing labels and separate successful from dead-end trajectories.

Interpretation

Figure note 5.3

This is an important validation because it moves beyond normal development. The ground truth comes from lineage tracing rather than only marker-gene knowledge, so it supports the claim that CellRank can recover real fate outcomes in perturbation settings.

5.4 Comparison with competing methods

The paper compares CellRank with Palantir, STEMNET, FateID, and velocyto. On pancreas data, only CellRank identifies both initial and terminal states automatically. Palantir can compute fate probabilities but requires an initial state and identifies fewer terminal states. STEMNET and FateID do not identify initial or terminal states, and velocyto does not provide fate probabilities.

The runtime and memory comparison shows that CellRank scales well to 100,000 cells. It is slower than STEMNET for fate probabilities but much faster and more memory efficient than Palantir and FateID in the displayed benchmark.

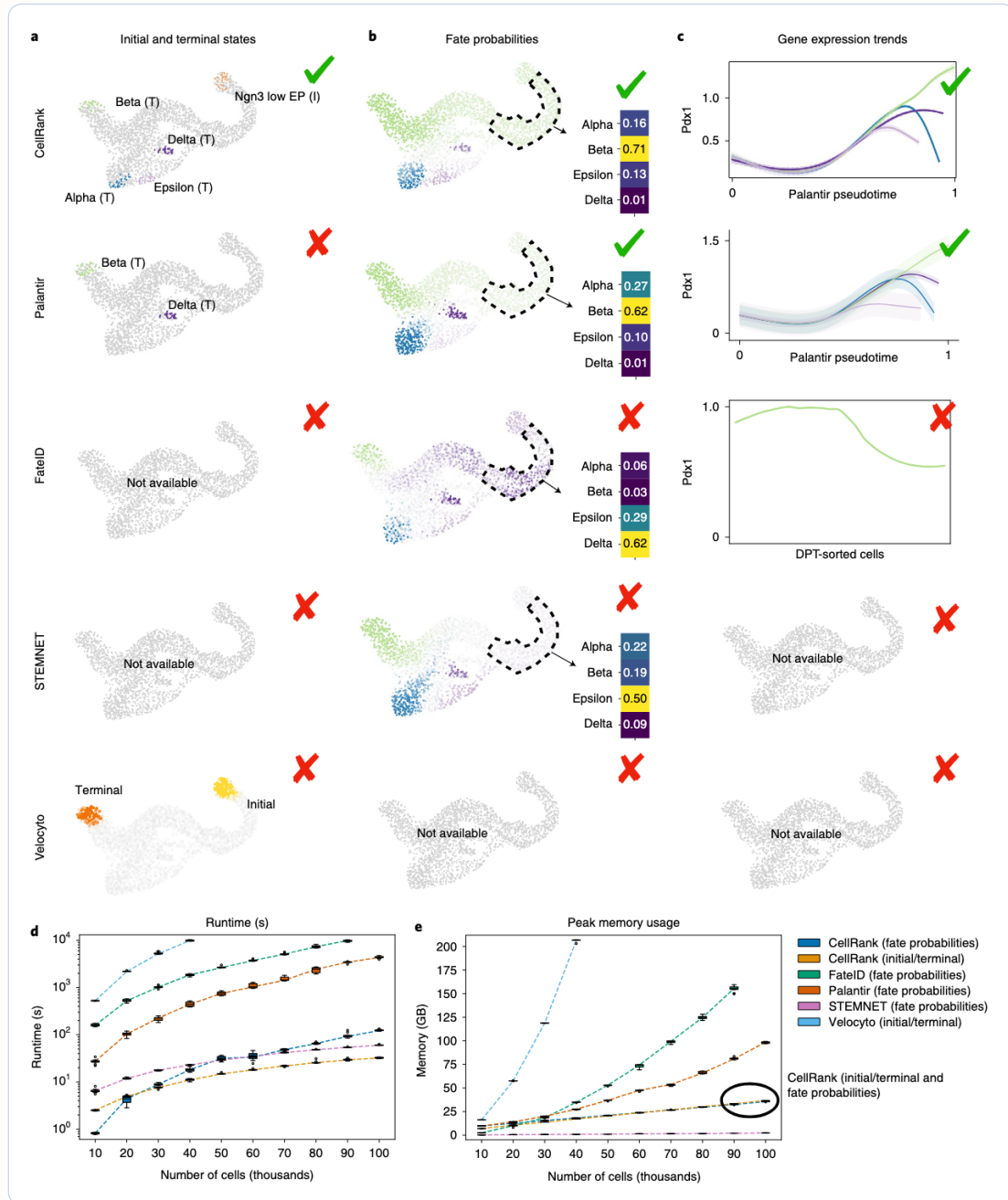


Figure 5: Main text Fig. 5a–e. Benchmark comparison. CellRank is compared with Palantir, FateID, STEMNET, and velocity for initial/terminal state detection, fate probabilities, gene-expression trends, runtime, and memory usage.

How to interpret the benchmark

Figure note 5.4

The key advantage of CellRank is not that it wins every computational metric. Its advantage is the combination of functions: automatic state discovery, probabilistic fate mapping, lineage-specific gene trends, and practical scalability.

5.5 Lung regeneration and dedifferentiation

The final biological application studies murine lung regeneration after bleomycin injury. This setting is different from normal development because cells may dedifferentiate rather than simply progress toward more mature states.

CellRank identifies airway macrostates and assigns high multilineage potential to MHC-II-positive club cells. More importantly, it predicts a goblet-to-basal dedifferentiation trajectory. The authors define three stages along this trajectory: goblet-like stage 1, intermediate stage 2, and basal-like stage 3. Stage 2 cells express both goblet and basal markers and are enriched after injury. The paper validates the predicted intermediate state using immunofluorescence staining.

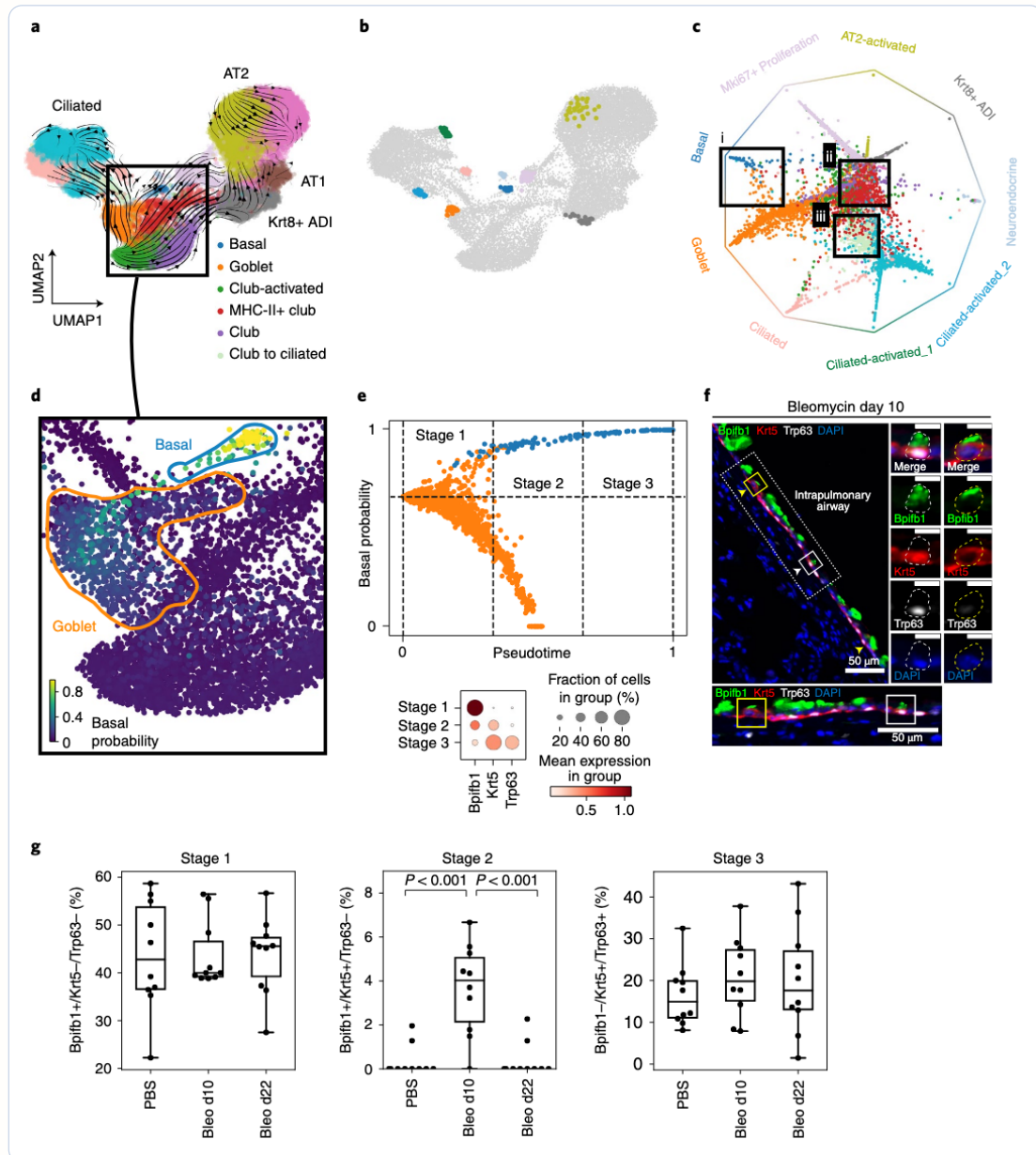


Figure 6: Main text Fig. 6a–g. Lung regeneration. CellRank predicts a goblet-to-basal dedifferentiation trajectory after injury and supports the existence of an intermediate Bpifb1-positive/Krt5-positive/Trp63-negative state.

Why this result is the strongest biological story

Figure note 5.5

The lung experiment is compelling because the predicted trajectory is unexpected and followed by experimental validation. However, the authors also note that the direction of the proposed trajectory still requires confirmation by lineage tracing.

6 Core mathematical intuition

6.1 Why a directed Markov chain?

A Markov chain is suitable because cell-state transitions are modeled as many small local steps. Each step is uncertain, but the aggregate of many steps can reveal a stable fate tendency. Directional information from RNA velocity turns an undirected similarity graph into a directed transition process.

From local to global

Intuition 6.1

RNA velocity says: among nearby cells, this one looks more like the near future. The Markov chain asks: if we repeatedly follow these local preferences, which terminal state will we eventually reach?

6.2 Why coarse-graining matters

Without coarse-graining, the transition matrix contains one state per cell and is difficult to interpret. GPCCA summarizes the system into macrostates while preserving the directional dynamics. Terminal states can then be defined through self-transition stability, and initial states through low revisit probability.

6.3 Why uncertainty propagation matters

RNA velocity can be noisy because unspliced RNA is sparse and affected by technical and biological noise. CellRank propagates velocity uncertainty into transition probabilities. In noisy regions, uncertain velocity vectors are downweighted, which makes fate probabilities more robust.

Method note

The uncertainty mechanism is important because it prevents the model from treating every velocity arrow as equally reliable. This is especially relevant in rare lineages or datasets with weak splicing signal.

7 Strengths of the paper

- **Strong conceptual integration:** CellRank combines pseudotime-style manifold assumptions with RNA-velocity-based directionality.
- **Probabilistic output:** fate probabilities are more biologically realistic than hard branch labels.
- **Automatic state discovery:** the method can infer initial, intermediate, and terminal populations instead of requiring all of them from the user.
- **High-dimensional analysis:** the transition matrix is computed in expression space rather than only on UMAP/t-SNE arrows.
- **Multiple biological settings:** the paper tests development, reprogramming, and regeneration.
- **Interpretability:** fate probabilities can be linked to lineage marker trends and candidate driver genes.

- **Experimental validation:** the lung regeneration case includes immunofluorescence validation of a predicted intermediate state.

8 Limitations and critical reading

Critical reading

CellRank is powerful, but its conclusions are only as reliable as the directional signal. If RNA velocity is systematically biased, fate probabilities can inherit that bias. Uncertainty propagation helps with noisy vectors, but it cannot fully correct a wrong vector field.

8.1 Dependence on RNA velocity quality

RNA velocity depends on spliced/unspliced mRNA modeling. Some genes or cell states may not have sufficient unspliced signal. This is especially problematic for rare fates, late-emerging populations, or systems where splicing kinetics do not match model assumptions.

8.2 Direction is inferred, not directly observed

CellRank predicts likely transitions, but it does not prove lineage ancestry. The lung goblet-to-basal trajectory is biologically interesting, but the paper explicitly states that direction still needs lineage-tracing confirmation.

8.3 Parameter and annotation dependence

Although the paper reports robustness to many parameters, choices such as number of neighbors, number of macrostates, velocity model, and terminal-state thresholds can still affect results. Macrostate annotation also depends on overlap with known clusters and marker genes.

8.4 Rare lineages remain challenging

The delta-cell example is both a strength and a limitation. CellRank can extract useful signal for rare lineages, but delta cells were not automatically classified as terminal under the default stability criterion. This suggests that manual biological judgment is still needed.

9 What I can reuse for my own research

Implications for my work

For pathology-omics and gene-regulatory-network modeling, the most reusable idea is to convert local directional evidence into global fate or phenotype probabilities. This could be adapted from cell-state transitions to disease-state transitions, tumor progression, or perturbation-response trajectories.

Potential reusable ideas:

- **Probabilistic fate output:** Instead of assigning each cell or tissue region to one endpoint, estimate probabilities toward multiple phenotypic endpoints.
- **Directed graph on a manifold:** Build transitions only between biologically similar states, then inject direction from a learned vector field.
- **Coarse-grained macrostates:** Summarize many cells or regions into interpretable states before analyzing dynamics.

- **Driver discovery through fate correlation:** Rank genes by their correlation with endpoint probabilities, then analyze temporal or pseudotemporal trends.
- **Uncertainty-aware transition modeling:** Propagate uncertainty from local predictions into downstream fate estimates.

10 Possible extensions

10.1 Multi-omics CellRank

The authors mention that other directional information could be used. A natural extension is to integrate chromatin accessibility, perturbation response, protein abundance, or metabolic labeling. For example, epigenetic changes often precede transcriptional changes, so chromatin-to-RNA delay could provide directionality.

10.2 Perturbation-aware CellRank

For Perturb-seq or CRISPR-screen data, a directed transition matrix could model how genetic perturbations move cells toward or away from disease phenotypes. Fate probabilities could then become treatment-response probabilities.

10.3 Pathology-omics adaptation

In computational pathology, one could imagine constructing a graph over cells, patches, or tissue regions. A learned direction could represent disease progression, tumor-stroma transition, immune exclusion, or therapy-induced state change. The CellRank idea would then become a probabilistic map of pathological phenotype trajectories.

11 Questions to ask while reading

Questions for further thought

1. If RNA velocity and marker-gene evidence disagree, which one should be trusted?
2. How sensitive are fate probabilities to the chosen velocity model?
3. Can terminal states be reliably detected in datasets without clear endpoint populations?
4. How should we validate inferred direction when lineage tracing is unavailable?
5. Can fate probabilities be used as supervision targets for downstream predictive models?

12 My final evaluation

Table 1: Personal evaluation of the paper.

Criterion	Evaluation
Novelty	High. The paper turns RNA velocity into a global probabilistic fate-mapping framework through directed Markov chains and GPCCA.
Biological value	High. The method produces interpretable fate maps and candidate drivers across development, reprogramming, and regeneration.
Technical clarity	Strong. The workflow is modular: KNN graph, velocity-informed transition matrix, macrostate coarse-graining, fate probabilities, gene trends.
Validation	Strong overall, especially because reprogramming uses lineage-tracing labels and lung regeneration includes experimental validation.
Main caveat	The method depends on the quality of the underlying velocity or directional vector field. Wrong directional signal can lead to wrong fate probabilities.
Reading rating	★ 4.7/5

Key takeaway

CellRank is best understood as a bridge between local directional estimates and global fate interpretation. Its most important output is not the velocity vector itself, but the probabilistic map of where each cell is likely to end up.